

Tongtong Wu

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Research Interests

Continual Learning: Focused on enhancing model capabilities to acquire new tasks while retaining previously learned information, aiming to mitigate catastrophic forgetting.

Natural Language Processing: Focused on generative information extraction using pretrained language models, especially in data-limited scenarios.

Knowledge Graph: Focused on knowledge graph construction and knowledge-augmented methods for downstream applications such as question answering.

For more detailed information on my research topics, please visit my Google Scholar profile.

Education

School of Computer Science and Engineering, Southeast University

Nanjing, China

PH.D. IN SOFTWARE ENGINEERING (AWARDED DEC 20, 2023)

Mar. 2018 - Dec. 2023

- Supervisor: Prof. Guilin Qi [gqi@seu.edu.cn]

Faculty of Information Technology, Monash University

Melbourne, Australia

VISITING PH.D STUDENT

Jan. 2020 - Nov. 2021

- Supervisor: Prof. Gholamreza Haffari [gholamreza.haffari@monash.edu]
and Prof. Yuan-Fang Li [yuanfang.li@monash.edu]

School of Computer Science and Engineering, Southeast University

Nanjing, China

M.S. IN SOFTWARE ENGINEERING

Jun. 2016 - Mar. 2018

- Supervisor: Prof. Guilin Qi [gqi@seu.edu.cn]

School of Internet of Things, Hohai University

Nanjing, China

B.S. IN INTERNET OF THINGS

Sep. 2012 - Jun. 2016

- Supervisor: Prof. Ning Sun [ningsun@hhuc.edu.cn]

Academic Positions

Faculty of Information Technology, Monash University

Melbourne, Australia

RESEARCH FELLOW (POSTDOC)

Oct. 2022 - Present

- Working with: Prof. Reza Haffari [gholamreza.haffari@monash.edu]

Faculty of Information Technology, Monash University

Melbourne, Australia

RESEARCH ASSISTANT ON CONTINUAL LEARNING AND LOW-RESOURCE NLP

Mar. 2021 - Jun. 2021

- Working with: Prof. Reza Haffari [gholamreza.haffari@monash.edu]

Gamma AI Lab, OneConnect Smart Technology, Ping-An Group

Shanghai, China

RESEARCH ASSISTANT (INTERNSHIP) ON FEW-SHOT LEARNING

Jan. 2019 - May. 2019

- Working with: Dr. Yujin Zhu [zhuyujin224@oneconnect.com]

School of Computer Science and Engineering, Southeast University

Nanjing, China

TEACHING ASSISTANT (DISCRETE MATHEMATICS)

Feb. 2017 - Jun. 2017

- Working with Prof. Guilin Qi [gqi@seu.edu.cn]

Teaching Experience

FIT5201 - Machine Learning, Monash University

Melbourne, Australia

ROLE: GUEST LECTURER

Semester 2, 2025

- Clustering algorithms (k-means) and latent variable models (Gaussian mixture model)
- Expectation-maximization algorithms

Supervision & Mentoring

PhD Students

ROLE: CO-SUPERVISOR

- [2025 - Present] **Shenyu Zhang** @ Monash University (LLM-based Memory Systems)
- [2024 - Present] **Rihui Jin** @ Monash University (Tabular and Textual Reasoning)
- [2023 - Present] **Peiwen Zhao** @ Beijing Institute of Technology (Temporal Reasoning with LLM)

Master's Students

ROLE: CO-SUPERVISOR

- [2025 - Present] **Kexin Dai** @ Monash University (Agentic Recommendation Systems)
- [2025 - Present] **Nipun Nair** @ Monash University (Conversational Recommendation Systems)
- [2024 - Present] **Jiazhen Kang** @ Southeast University & Monash University (Retrieval-augmentation Code Generation)
- [2023 - Present] **Rongyi Chen** @ Southeast University & Monash University (Configurable Code Agent)

Alumni

ROLE: CO-SUPERVISOR

- [2022 - 2025] **Guitao Wang** Mphil @ Southeast University (Speech Continual Learning) -> Phd @ Monash University
- [2021 - 2024] **Jingqi Kang** Mphil @ Southeast University (Speech Information Extraction) -> ML Engineer @ iFLYTEK

Research Projects & Grants

Lifelong Version-controlled Code Generation: Lifelong Version-controlled Code Generation

AUD 170K

REZA HAFARI (PCI), **TONGTONG WU** (CI)

07/2025 - 07/2026

- eBay Research Grant

Towards Sustainable Software Engineering Multi-Agent Systems

AUD 30K

XIAONING DU (PCI), **TONGTONG WU** (CI)

08/2025 - 08/2026

- Monash Seed Grant

Continual Learning of Large Language Models for Version-Controlled Code Generation

AUD 32K

TRANG VU (PCI), **TONGTONG WU** (CI)

08/2024 - 08/2025

- Monash Seed Grant

Computational Cultural Understanding (CCU)

AUD 4M

RESEARCH FELLOW (FULL TIME)

10/2022 - 07/2024

- DARPA Project, working with Prof. Reza Haffari

Large-scale Multi-modal Knowledge Management (KMASS)

AUD 3M

RESEARCH FELLOW (PART TIME)

10/2022 - 02/2023

- DARPA Project, working with Prof. Dragan Gasevic

Selected Publications

K-DeCore: Facilitating Knowledge Transfer in Continual Structured Knowledge Reasoning via Knowledge Decoupling

NeurIPS

YONGRUI CHEN, YI HUANG, Y. LIU, SHENYU ZHANG, JUNHAO HE, **TONGTONG WU**, GUILIN QI, TIANXING WU

2025

- Core Rank - A*

Can LLMs Evaluate Complex Attribution in QA? Automatic Benchmarking Using Knowledge Graphs

ACL

NAN HU, JIAOYAN CHEN, Y. WU, GUILIN QI, H. WANG, SHENG BI, YONGRUI CHEN, **TONGTONG WU**, JEFF Z PAN

2025

- Core Rank - A*

Continual Speech Learning with Fused Speech Features

Interspeech

GUITAO WANG, JINMING ZHAO, HAO YANG, GUILIN QI, **TONGTONG WU***, GHOLAMREZA HAFFARI

2025

- Core Rank - A, * Corresponding Author

Continual Multimodal Knowledge Graph Construction

IJCAI

XIANG CHEN, JINTIAN ZHANG, XIAOHAN WANG, NINGYU ZHANG, **TONGTONG WU**, CHENGMING WANG, FEIYU XIONG,

WEI GUO AND HUAJUN CHEN

2024

- Core Rank - A*

Towards LifeSpan Cognitive Systems

Trans. Mach. Learn. Res.

YU WANG*, CHI HAN*, **TONGTONG WU***, XIAOXIN HE*, WANGCHUNSHU ZHOU, NAFIS SADEQ, XIUSI CHEN, ZEXUE

HE, WEI WANG, GHOLAMREZA HAFFARI, HENG JI, JULIAN MCAULEY

2024

- * Equal Contribution

KCG-EE: Effective Knowledge Conditioning for Generation-based Event Extraction

World Wide Web J.

TONGTONG WU, FATEMEH SHIRI, JINGQI KANG, GUILIN QI, YUAN-FANG LI, REZA HAFFARI

2023

- Core Rank - A

Pretrained Language Model in Continual Learning: A Comparative Study

ICLR

TONGTONG WU, MASSIMO CACCIA, ZHUANG LI, YUAN-FANG LI, GUILIN QI, REZA HAFFARI

2022

- Core Rank - A*

Towards Relation Extraction From Speech

EMNLP

TONGTONG WU*, GUITAO WANG*, JINMING ZHAO*, Z. LIU, GUILIN QI, YUAN-FANG LI, GHOLAMREZA HAFFARI

2022

- Core Rank - A*, * Equal Contribution

Curriculum-Meta Learning for Order-robust Continual Relation Extraction

AAAI

TONGTONG WU, XUEKAI LI, YUAN-FANG LI, REZA HAFFARI, GUILIN QI, YUJIN ZHU, GUOQIANG XU

2021

- Core Rank - A*

Selected Under Review

VersiCode: Towards Version-controllable Code Generation

TONGTONG WU, WEIGANG WU, XINGYU WANG, KANG XU, SUYU MA, BO JIANG, PING YANG, ET.AL

- New Benchmark

Environment-Aware Code Generation: How far are We?

TONGTONG WU, RONGYI CHEN, WENJIE DU, SUYU MA, GUILIN QI, ZHENCHANG XING, SHAHRAM KHADIVI, RAMESH

PERIYATHAMBI, GHOLAMREZA HAFFARI

- Empirical Study, Conditional Accepted by ICSE 2026, Core Rank - A*

Structuring API Evolution: A Graph-Guided Framework for LLM-Based Code Migration

TONGTONG WU, JIAZHEN KANG, YUCHEN LU, GUILIN QI, SUYU MA, KANG XU, NINGYUAN SUN, BO JIANG, ZHENCHANG XING, GHOLAMREZA HAFFARI

- New Method

CARD: Towards Conditional Design of Multi-agent Topological Structures

TONGTONG WU, YANMING LI, ZIYE TANG, CHEN JIANG, LINHAO LUO, GUILIN QI, SHIRUI PAN, GHOLAMREZA HAFFARI

- New Method

Selected Patents

A Document-level Event Extraction Method Based On Task-conditioned Generation

Chinese Patent

TONGTONG WU, JINGQI KANG, GUILIN QI

- Granted (202210403218.6)

2022

A Continual Relation Extraction Method Based on Task-aware Meta Learning

Chinese Patent

TONGTONG WU, XUEKAI LI, GUILIN QI

- Granted (202110099305.2)

2021

Professional Service

Member of Program Committee

SELECTED CONFERENCES

• ICML [2023 - 2025], NeurIPS [2023 - 2025], ICLR [2023 - 2026], ACL ARR [2022 - 2025], AAAI [2022, 2024 - 2026], IJCAI [2023 - 2024]

Reviewer of Journals

SELECTED JOURNALS

- TKDE (IEEE Trans. Knowl. Data Eng.), TCSS (IEEE Trans. Comput. Soc. Syst.)
- TCSVT (IEEE Trans. Circuits Syst. Video Technol.), TIP (IEEE Trans. Image Process.)
- TIST (ACM Trans. Intell. Syst. Technol.), TALLIP (ACM Trans. Asian Low Resour. Lang. Inf. Process.), KBS (Knowl. Based Syst.)

Editorial Board Member of Journals

SELECTED JOURNALS

- DI (Data Intelligence) [IF - 2.31]

Awards & Scholarships

Jun. 2020 **Chinese Government**, Chinese Scholarship Council Scholarship

Beijing, China

Sep. 2018 **Southeast University**, Outstanding Postgraduates

Nanjing, China

Jun. 2016 **Hohai University**, Excellent Graduates - Best 50

Nanjing, China

Sep. 2014 **Hohai University**, Outstanding Student Pacesetter

Nanjing, China

Sep. 2013 **Chinese Government**, National Encouragement Scholarship

Nanjing, China

Community Engagement

Nov 2025 **Continual Learning of Large Language Models**, (3-hr) Tutorial Speaker @ EMNLP

Suzhou, China

Nov 2024 **Continual Learning of Large Language Models**, (3-hr) Tutorial Speaker @ AJCAI

Melb., Australia

Oct 2024 **The Advances of Continual Learning with LLMs**, Keynote Speaker of CL workshop @ ACM-MM

Melb., Australia

Aug 2024 **Towards Version-Controllable Code Generation**, Invited Speaker @ Oracle AI Research Seminar

Melb., Australia

Mar 2022 **Continual Learning with Pretrained Language Models**, Invited Speaker @ AI-TIME

LIVE Online